

Emre Cecanpunar

C and modern C++ (C++17–23) systems engineer focused on performance-critical software:
low-latency and high-throughput systems, compilers, drivers, and OS internals.

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Open Source Contributions

Mesa

2025 – Present

GPU compilers, Vulkan drivers, and performance analysis

- Develop and debug C and C++17 code across Mesa shader compilers and Vulkan drivers, from shader IR and code generation to allocation lifetime and hardware-facing error paths.
- Profile Intel's BRW compiler with Linux perf and full shader-db; a [scoreboard optimization](#) improved compiler throughput by cutting task-clock 6.16% and the hot path's share of P-core cycles from 5.10% to 1.42%, with identical output across 26,884 stable shader-db lines.
- Test changes on Intel graphics (ANV) and Arm Mali-G610 (Panfrost) hardware using targeted Vulkan reproducers, Vulkan CTS, IR and generated-code inspection, and ASan/UBSan builds.

Linux kernel

2025 – Present

Platform drivers, DRM, and low-level systems

- Contribute focused C fixes across platform/x86 and DRM drivers, including [hp-wmi](#), [i915](#), and [Panthor](#).
- Diagnose memory-safety, concurrency, undefined-behavior, and error-path issues in hardware-facing kernel code using ftrace function-graph traces, trace-cmd, KGDB, and KASAN/UBSan builds.

LLVM

2026 – Present

MC/ARM

- [Merged a fix](#) to preserve label-time ISA state for Thumb function symbols when a later `.type` directive marks the function.

Apache NuttX

2026

Arm64 SoC bring-up and upstream fixes

- [Upstreamed initial NuttX support](#) for the Rockchip RK3588S SoC and Orange Pi 5 Pro, booting through U-Boot to NSH with GICv3, timer, UART, and PSCI support. [Project announcement](#).
- [Fixed two EDID parsing bugs](#): incorrect standard-timing decoding and use of the wrong chromaticity macro.

Other upstream work

2026

- [Merged a heap-buffer-overflow fix](#) in NVIDIA egl-wayland2 and [submitted a use-after-free fix](#) in Wayland.

Experience

Embedded Software Engineer Intern, Baykar Technologies

Feb 2025 – Jun 2025

T3 Gemstone platform

- Initiated the [Apache NuttX](#) port to the TI AM67A SoC: brought up an Arm Cortex-R5F core, configured the MPU and caches, defined the memory map, and implemented the early boot path.
- Wrote linker scripts, GPIO drivers, and UART drivers; debugged NuttX ELF loading through Linux remoteproc using readelf to inspect load addresses, sections, vectors, and resource-table requirements.

Avionics Software Developer, Atmaca Rocket Team

Sep 2022 – May 2023

University rocketry team · TÜBİTAK competition

- Developed redundant-controller avionics firmware and a parachute-deployment algorithm using Kalman-filtered multi-sensor altitude estimates; the avionics technical report received a full score from TÜBİTAK.

Education

B.Sc. Computer Science & Engineering, Akdeniz University

Sep 2022 – Jun 2026